



Indian Institute of Technology, Kanpur
भारतीय प्रौद्योगिकी संस्थान कानपुर



Department of Electrical Engineering

Placement Brochure



www.iitk.ac.in

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About the Department

History

The Electrical Engineering Department was one of the first few departments with which IIT Kanpur started in 1960. Advanced Centre for Electronics Systems (ACES) came into existence in early 1970 by a grant from the Ministry of Defence to undertake manpower training and to carry out research and development related to defence. Subsequently, several sponsored projects in many subject areas with large interdisciplinary content were handled through ACES.

ACES is now an integral part of Electrical Engineering Department and houses majority of Electrical Engineering Laboratories and Facilities. Electrical Engineering Department is currently the largest multidisciplinary department at IIT Kanpur and covers practically all sub-disciplines under Electrical Power, Electronics and Communications.

60+ Faculties

50+ Patents

100+ Research Projects

Programs Offered

Undergraduate Programs

- B.Tech. in Electrical Engineering: 4-year foundational degree.
- Double Major: Secondary full major option for IITK students.
- Minor in EE: Specialized technical track for non-EE students.

Postgraduate & Research

- M.Tech.: 2-year program blending coursework and advanced research.
- MS (Research): 2-year thesis-oriented degree focused on industry research projects.
- Ph.D.: Cutting-edge doctoral research with global joint-degree pathways.

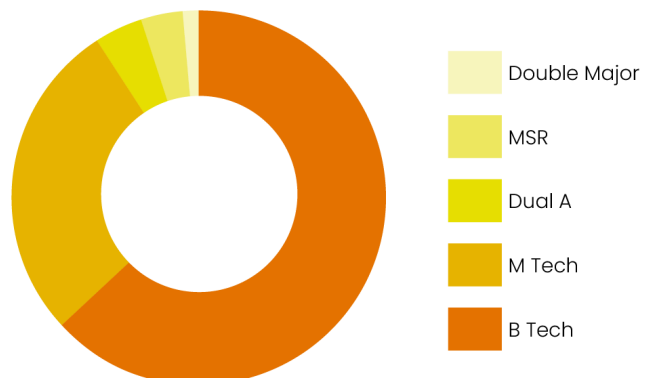
Dual & Professional Tracks

- B.Tech.-M.Tech. Dual Degree: Integrated 5-year academic fast-track.
- Online M.Tech.: Flexible industry-aligned degrees for professionals.

Current Student Demography.

A total of 220 students are eligible to participate in the campus placement process across multiple academic programs. The eligible cohort includes 139 B.Tech students, 61 M.Tech students, 9 Dual A students, 8 MSR students, and 3 Double Major students.

This diverse pool brings together students from varied academic backgrounds and areas of specialization, providing recruiters with access to a broad and multidisciplinary talent base for their hiring requirements.



Message from the **HOD**



Dear Recruiters,

It is with great pleasure that I welcome you on behalf of the Department of Electrical Engineering at the Indian Institute of Technology Kanpur. As the Head of the Department, I am proud to introduce our exceptionally talented students, who embody academic excellence, innovation, and industry readiness.

With a legacy spanning over six decades, the Department of Electrical Engineering has been a pioneering part of the institute since its inception in 1960. Today, the department stands at the forefront of research and development, supported by cutting-edge facilities such as RTDS, Smart Grid, 5G Test Bed, EMI-EMC Testing, and Antenna and Semiconductor Device Measurement facilities, among many others. Our academic and research activities span traditional areas including Power Engineering, Control Systems, Signal Processing, Microelectronics, and VLSI, as well as emerging fields such as Artificial Intelligence, Machine Learning, Neuromorphic Computing, Hardware Security, and Future Wireless Communication Networks, including 5G, 6G, and mmWave. Our students have developed strong expertise across these diverse technological domains.

We maintain a strong connection between academia and industry through research projects and active student involvement. Strategic partnerships with industry leaders provide valuable opportunities for hands-on experience, real-world projects, and internships, giving our students a distinct competitive edge. This is reflected in our placement record and the strong demand for our graduates among leading companies globally. Backed by a solid technical foundation, analytical ability, and practical readiness, our students are prepared to contribute meaningfully from day one.

I warmly invite you to discover the exceptional potential and talent of our Electrical Engineering students. They are innovators, critical thinkers, and future leaders, prepared to make substantial contributions to your organization. We look forward to building strong and mutually beneficial relationships with your esteemed organizations.

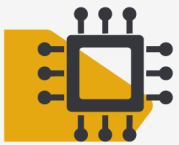
**Best wishes,
Prof. Yogesh Singh Chauhan
Head EE**



**Email: head_ee@iitk.ac.in
Phone: 0512-259-7244**

Academic & Research Strengths

The Department of Electrical Engineering at IIT Kanpur combines rigorous academic training with cutting-edge research in Microelectronics & VLSI, RF & Microwaves, Signal Processing & Communications, Power Engineering, Control & Automation, and Optoelectronics & Optical Communication. Our work spans emerging technologies such as 5G/6G and mmWave systems, RFICs, advanced semiconductor devices, neuromorphic computing, AI, robotics, power electronics, quantum technologies, and integrated photonics. Supported by advanced laboratories and extensive hands-on research, our students develop strong analytical, design, and problem-solving skills to translate fundamental concepts into real-world technologies.



Microelectronics & VLSI

The Microelectronics & VLSI specialization covers research across Microelectronics, VLSI System Design, Analog and Digital Circuit Design, Semiconductor Device Modelling and Simulation, Solid-State Devices, Nanoelectronics and Nano-scale Devices, Organic Electronics, Flexible Electronics, and Photovoltaics.



RF and Microwaves

Research covers Electromagnetics, RF Engineering & Microwaves, Antennas, Metamaterials, MMICs, RF & Microwave Sensors, RFID, Microwave & Millimeter-wave Imaging, RF Energy Harvesting, Electromagnetic & Tomographic Imaging, and Terahertz Imaging & Testing.



Signal Processing, Communications & Networks

Research areas include Digital Communication Systems, Information & Coding Theory, Telecom & Wireless Networks, Peer-to-Peer Networks, Digital Switching Systems, 5G Wireless Communication Systems, Digital Signal & Image Processing, Computer Vision, Inverse Problems & Tomography, and Signals & Systems Theory



Power Engineering

Power Systems Economics; Optimization and Markets; Power System Protection; HVDC & FACTS, Power Quality; Smart Grid and Synchro phasors; Power Electronics; Electric Drives; Active Power Filters and Static VAR Systems; Renewable Energy Interfaces.



Control & Automation

Research encompasses Control & Dynamical Systems, Robotics, Networked Control, Electric Vehicle Control, Electronic & Virtual Instrumentation, Fuzzy Logic, and Neural Networks & their Applications.



Optoelectronics & Optical Communication

Research includes Nanophotonic, Plasmonic & Quantum-Dot Devices, Optoelectronics, Signal Processing for Fibre Optics, Nonlinear Fibre Optics, Fibre-Optic Sensors, Quantum Cryptography & Quantum Optics, Spin Waves, and Photonic Networks & Systems.

Labs & Facilities

The department's infrastructure enables a seamless integration of theoretical learning, experimental research and technology development, giving students exposure to sophisticated tools and platforms used in contemporary electrical and electronics engineering.

Microelectronics & VLSI

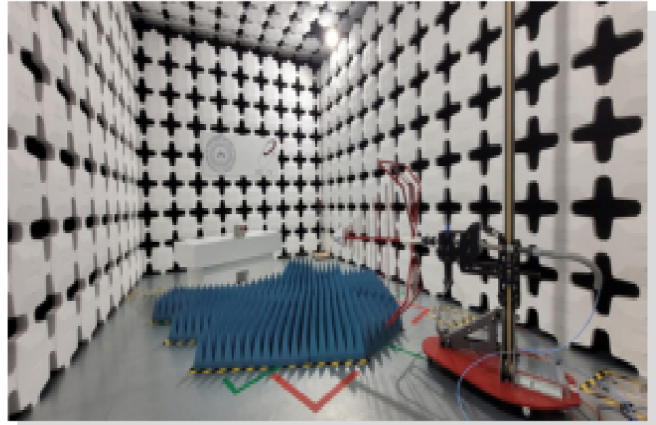
- Microelectronics, VLSI and Display Technologies Facilities
- Semiconductor Device Fabrication Laboratory
- VLSI Laboratory
- Organic Electronics Laboratory
- Electronic Circuits Laboratory
- Microprocessor and Microcontroller Laboratory

RF & Microwave

- RF, Microwave and Photonics Facilities
- Microwave Circuits and Measurement Laboratory
- Microwave Imaging and Material Testing (MIMT) Laboratory
- Antennas Laboratory
- Anechoic Chamber
- RFID Laboratory
- Electromagnetic Compatibility/Interference Laboratory
- PCB Fabrication Facility

Signal Processing, Communication & Networks

- Signal Processing, Communication & Networks Facilities
- Digital Signal Processing Laboratory
- Computer Vision II Laboratory
- Mobile Communications
- Multimedia Wireless Networks Laboratory



- Multimodal Information Processing Systems Laboratory
- Networks Laboratory
- Wireless Communications Coding and Cognitive Radio Laboratory (WC3 Laboratory)

Power & Energy

- Power Engineering Facilities
- High Voltage Laboratory
- Electro Mechanical Energy Conversion (EMEC) Laboratory
- Power Electronics Laboratory
- Power Management Laboratory
- Power System Simulation and Research Laboratory
- Static Controller Laboratory

Control & Automation

- Control and Automation Facilities
- Control Systems Laboratory
- Networked Control Systems Laboratory
- Intelligent Systems Laboratory
- Data Mining Laboratory
- Computer Vision Laboratory
- Computer Vision Laboratory II
- Intelligent Health Monitoring Systems Laboratory
- Bioinformatics Laboratory

Optoelectronics & Photonics

- Fibre and Quantum Optics Laboratory
- Optoelectronics and Nanofabrication Laboratory



Course Curriculum

Microelectronics & VLSI

EE610: Analog/Digital VLSI Circuits
 EE611A: Organic Electronics
 EE613: High Frequency Analog Circuit Design
 EE614: Solid State Devices
 EE615A: Solar Photovoltaic Technologies
 EE616A: Semiconductor Device Modelling
 EE618: Integrated Circuit Fabrication Technology
 EE619: VLSI System Design
 EE681A: Compact Modelling
 EE698D: Special Topics in Microelectronics
 EE698G: Circuits for Phase and Frequency Synthesis
 EE698I: Mixed-Signal Integrated Circuit Design
 EE698L: RFIC Design
 EE698N: Introduction to Flexible Electronics
 EE698P: Memory Technology and Neuromorphic Computing
 EE698W: Analog Circuits for Signal Processing

Signal Processing, Communications & Networks

EE602: Statistical Signal Processing
 EE604: Image Processing
 EE605: Introduction to Signal Analysis
 EE621: Representation and Analysis of Random Signals
 EE623: Detection and Estimation Theory
 EE627: Speech Signal Processing
 EE667: Information Theory
 EE670: Wireless Communications
 EE675: Introduction to Reinforcement Learning
 EE677: MIMO Wireless Communications
 EE698K: Programming for Signal Processing
 EE698Q: 5G Wireless Technologies
 EE708: Fundamentals of Data Science and Machine Intelligence
 EE798L: Machine Learning for Wireless Communications

RF & Microwaves

EE638: Advanced RF Antennas
 EE640: Computational Electromagnetics
 EE641: Advanced Engineering Electromagnetics
 EE642: Antenna Analysis & Synthesis
 EE643: Smart Antennas for Mobile Communications
 EE645: Monolithic Microwave ICs
 EE647: Microwave Measurements and Design

EE648: Microwave Circuits

EE649: The Finite Element Method for Electric and Magnetic Fields

Power Engineering

EE630: Simulation of Modern Power Systems
 EE631: Advanced Power System Stability
 EE635: HVDC Transmission & Flexible AC Transmission Systems
 EE660: Basics of Power Electronic Converters
 EE662: Control Techniques in Power Electronics
 EE664: Fundamentals of Electric Drives
 EE698B: Smart Grid Technologies Applications
 EE698E: Power Converters for EV Charging
 EE698H: Operation and Control of the Power System with Inverter-Based Resources

Control & Automation

EE637: Mathematical Methods in Control Systems
 EE650: Basics of Modern Control Systems
 EE651: Nonlinear Systems
 EE652: Linear Stochastic Dynamical Systems
 EE653: Digital Control
 EE654: Robust Control Systems
 EE655: Optimal Control
 EE656: Control System Design
 EE666: Special Topics in Power Electronics
 EE678: Neural Systems and Networks

Optoelectronics & Optical Communication

EE612: Fiber Optic Systems
 EE629: Digital Switching
 EE644A: Optical Communication
 EE646: Photonic Networks and Switching
 EE798I: Nanophotonics
 EE698E: Quantum and Wave Phenomena

Notable Recent Achievements

Student

- **Qualcomm Innovation Fellowship (QIF) 2025** — Mayank Anupam, Mohmad Aasif Bhat, Abhisha Garg, and Raghav Shukla.
- **IEEE Signal Processing Society (SPS) Scholarship 2025** — Abhisha Garg, Shivangi Dubey, and Hidangmayum Bebina Devi.
- **IEEE MTT-S Graduate Fellowship 2025** — Baisakhi Bandyopadhyay, awarded at the IEEE International Microwave Symposium.
- **IEEE AP-S C.J. Reddy Travel Grant** — Shreya Pourush.
- **European School of Antennas (ESoA) 2026 Course Grant** — Shreya Pourush, for Metalenses for Antenna Applications.
- **GRID-INDIA Power System Awards (GIPSA) 2025-26** — Jalaj Kumar.
- **Hyundai Hope Scholarship** — Abhishek Bhardwaj, for developing an AI-Driven Headset for Visually Impaired People.
- **Saransh Thesis Competition 2025, INYAS** — First Prize — Anil Kumar Panda.
- **Silver Medal for Excellence in Education, Sitaram Jindal Foundation** — Challagulla Baala Krishhna.
- **1st Place, Analog Tapeout Contest, VLSID 2026** — Mayank Anupam.

Alumni

- **Prof. Rajesh K. Gupta (B.Tech EE, 1984)** — Founding Dean, School of Computing, Information and Data Sciences, University of California San Diego.
- **Dr. Neeraj Mittal (B.Tech EE, 1989)** — Secretary, Ministry of Petroleum and Natural Gas, Government of India.
- **Prof. Ajit Kumar Chaturvedi (B.Tech-M.Tech/PhD EE, 1988/1995)** — Vice-Chancellor, Banaras Hindu University (BHU).
- **Dr. Ajay Kumar (B.Tech EE, 1984)** — Chairman, Union Public Service Commission (UPSC).
- **Mr. Sandeep Fuller (B.Tech EE, 1986)** — CEO — Systems & PMC, SYSTRA, an international engineering and infrastructure consultancy.
- **Prof. Rajiv K. Varma (B.Tech/PhD EE, 1980/1988)** — IEEE Fellow, recognizing his contributions to electrical engineering.
- **Mr. Amit Agrawal (B.Tech EE, 1991)** — Secretary, Department of Pharmaceuticals, Government of India.
- **Prof. Saikat Guha (B.Tech EE, 2002)** — IEEE Fellow, Class of 2025.
- **Dr. Pravesh Kothari (B.Tech EE, 2010)** — Presburger Award for contributions to theoretical computer science.
- **N. R. Narayana Murthy (M.Tech EE, 1969)** — Founder of Infosys and a pioneering technology entrepreneur who helped establish India as a major global software and IT services hub. IIT Kanpur recognizes his role in building an entrepreneurial venture that placed India on the international software map.
- **Dr. Girish Pahwa (PhD EE, 2020)** — Assistant Professor at National Yang Ming Chiao Tung University (NYCU), Taiwan, with research spanning semiconductor devices, device modelling, and emerging semiconductor technologies.

Placement Process



Past Recruiters




TEXAS
INSTRUMENTS



MIPS

Qualcomm

Flipkart



life.augmented

Rakuten

micron™



Samsung
Research

NVIDIA®

MEDIATEK

MAXLINEAR



MORPHING MACHINES

SIEMENS
energy



MathWorks®

Schlumberger

ananant
compute • connect • cognize



ESPRESSIF



Aura
Semiconductor



ENPHASE.



ceremorphic®



भारत इलेक्ट्रॉनिक्स
BHARAT ELECTRONICS

TVS



axiyo

NXP

SIEMENS



DELTA

publicis
sapient



Have Questions? **Contact Us**

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Departmental Placement Coordinators



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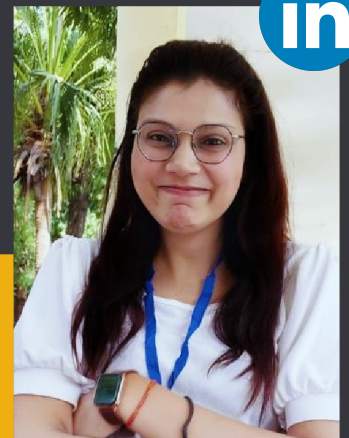
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